

Name	ID
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1.

Name	NID	Trait
Adam	200	Simple
Eve	100	Complex
Lilith	402	Toxic

Construct a hash index on the attribute “NID” of the above table. The hash index has 4 buckets, each capable of holding a maximum of 1 index entry. Bucket overflow is resolved using forward chaining. The hash function, $h = (\text{NID}) \% 100$

Then, explain if the hash function is a good uniform one or not.

[3]

2. Construct a B+ tree of order $n = 4$ for the following search key values inserted in the given order: 16, 18, 24, 8, 5, 10, 22, 30, 27, 17, 19, 28. Each time there is a split, a new B+ tree must be drawn. **[7]**

Name	ID
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1.

Name	NID	Trait
Adam	2101	Simple
Eve	1002	Complex
Lilith	4002	Toxic

Construct a hash index on the attribute “NID” of the above table. The hash index has 4 buckets, each capable of holding a maximum of 1 index entry. Bucket overflow is resolved using forward chaining. The hash function, $h = (\text{NID}) \% 1000$

Then, explain if the hash function is a good uniform one or not.

[3]

2. Construct a B+ tree of order $n = 4$ for the following search key values inserted in the given order: 16, 18, 24, 8, 5, 10, 22, 30, 27, 17, 19, 28. Each time there is a split, a new B+ tree must be drawn. **[7]**